How Much of a Blood-Brain-Barrier GNN's Performance Survives Honest Evaluation?

Nabil

Abstract

We a
extern
(7,805
reproc
featur
(over-
ECFP

1. Introduction

Predicting BBB permeability is core to CNS drug discovery. A metric is only as trustworthy as the protocol behind it. Random splits leak structural similarity (near-identical scaffolds on both sides); scaffold splitting controls this. 'External' validation leaks when the external set contains training compounds. We take one model reporting ~0.96 AUC and ask how much survives honest evaluation, and whether its architecture is justified. We do not claim to discover the leakage phenomenon (it is established); we contribute a controlled audit and a reusable, registry-driven harness.

2. Methodology

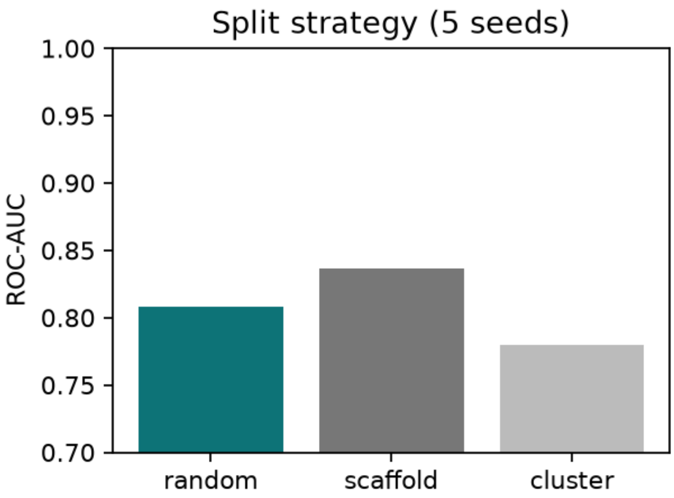
Data: BBBP (2,039 cpds, 76% BBB+); B3DB as external set. Only 31% of BBBP has a defined stereocentre. Featurisation: 21 node features (15 atomic + 6 stereo), 18 edge features. Model: GATv2 encoder (4 layers, hidden 128) + classifier; AdamW, BCE, early stopping on val AUC. Exp 2 uses the deployed pretrained 5-fold ensemble; Exp 1/3/5 retrain from scratch to isolate each variable. Splits: random, scaffold (Bemis-Murcko), cluster (KMeans on ECFP4); every model sees the same split per seed. Code is a small library with a model registry, so adding a model to the benchmark is one line.

3. Experiments and results

Exp 1: split strategy (5 seeds)

Split		
Random	0.897+/-.022	0.951
Scaffold	0.840+/-.026	0.938
Cluster	0.854+/-.064	0.921

Random-minus-scaffold gap = +0.057.



Exp 2: external validation (deployed ensemble)

Set				
Naive	7805	0.966	0.979	0.671
Dedup	6119	0.959	0.974	0.661

21.6% of B3DB overlaps BBBP training, yet removing it costs only 0.007 AUC. The 0.96 reproduces and is NOT exact-duplicate memorisation. High sensitivity with low specificity (0.66) foreshadows the calibration problem.

Exp 3: ablation (3 seeds)

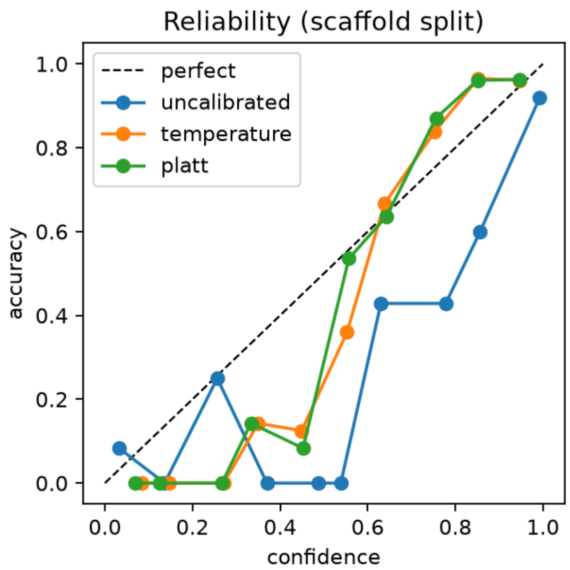
Stere		
Full	0.899	0.848
No stereo	0.908	0.868

Stereo features give no measurable benefit (within seed noise ~0.025). Expected: passive BBB diffusion is stereo-insensitive and ~2/3 of BBBP is achiral (carrier-mediated transport, e.g. L-DOPA via LAT1, can be stereoselective but is the minority route). Depth (scaffold): 4 layers 0.848, 2 layers 0.860, 1 layer 0.837 -> over-parameterised.

Exp 4: calibration (scaffold split)

Metho		
Uncalibrated	0.106	0.109
Temperature	0.080	0.101
Platt	0.070	0.100

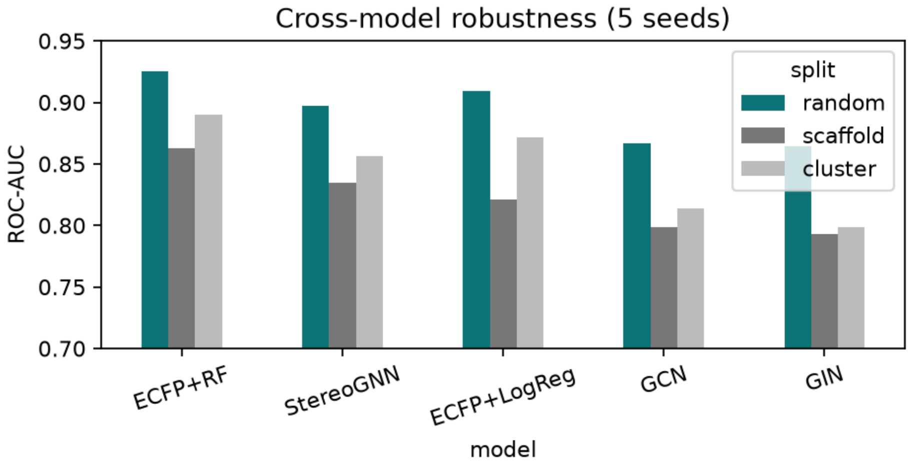
Fitted temperature T=4.30 (>1 = over-confident). Scaling halves the ECE.



Exp 5: cross-model benchmark (5 seeds)

Model			
ECFP+RF	0.926	0.863	0.890
StereoGNN	0.897	0.834	0.856
ECFP+LogReg	0.909	0.821	0.872
GCN	0.867	0.799	0.814
GIN	0.864	0.793	0.799

ECFP+RandomForest beats every GNN on every split, training in under a second. Your StereoGNN is the best of the neural models on scaffold.



4. Analysis and conclusion

The 0.96 reproduces and is not a duplicate-memorisation artefact, but it is a random-split-favourable number: scaffold splitting costs ~0.057, and the original 'beats 8 SOTAs' comparison was not like-for-like. AUC hides a positive bias (specificity 0.66) explained by over-confidence (T=4.30). The stereo features earn nothing on BBB (expected; they likely help on stereo-sensitive endpoints like transporter activity), so stereo should be an explicit optional mode. The model is over-parameterised, and a half-second ECFP+RF baseline beats it under honest evaluation. Evaluation design, not architecture, drove much of the headline. Deliverable: an honest characterisation plus a reusable harness that makes the audit one line of code per added model.